INCH-POUND

MIL-C-49285/45
19 October 1989

MILITARY SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, THREE INDIVIDUALLY SHIELDED PAIRS OF 24 AWG SOLID WIRE

NOTE: NOT FOR AEROSPACE USE.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49285.

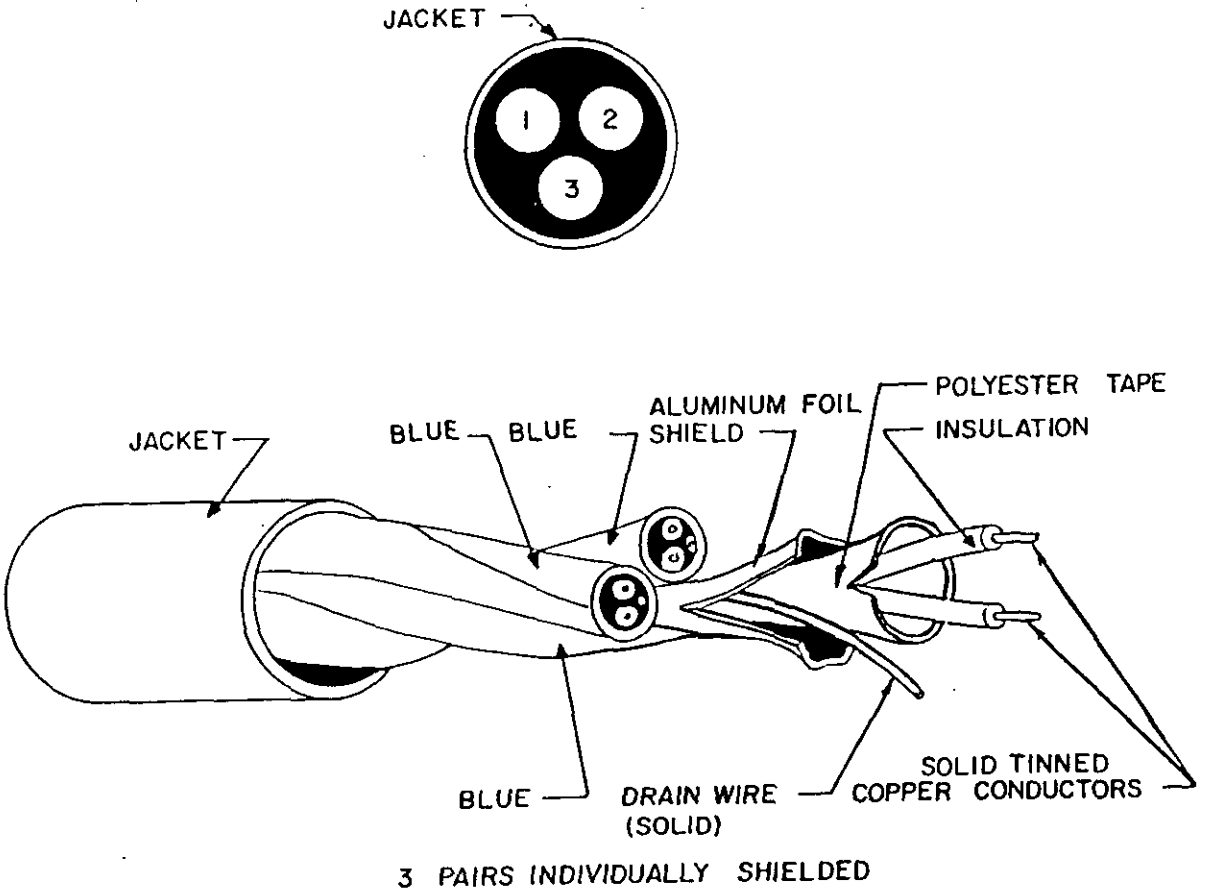


FIGURE 1. Cable illustration.

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TABLE I. Description, electrical.

Electrical characteristics	Requirements
Nominal capacitance between conductors	45 pF/ft (picofarads/foot)
Nominal capacitance between the conductor and the shield with one conductor tied to the shield	85 pF/ft
Continuous working voltage	300 V rms maximum between conductors
Nominal capacitance between adjacent shields	150 pF/ft
Continuous working voltage between adjacent shields	50 volts maximum
Shield dc resistance	20.5 ohms/1,000 feet maximum

TABLE II. Description, physical.

Components	Construction details
Number of pairs	Three individually shielded
Conductor type and wire size	Solid tinned copper 24 AWG
Conductor insulation	Polyvinyl chloride
Conductor insulation thickness	0.010 inch nominal thickness
Drain wire type and size	Solid tinned copper 24 AWG
Jacket material	Polyvinyl chloride
Jacket thickness	0.030 inch
Finished cable diameter	0.275 inch maximum outside diameter
Cable style (UL)	2493
Tensile strength (jacket)	2,000 pounds per square inch minimum
Elongation (jacket)	150 percent minimum
Overall cabling lay lengths	3.75 twists per foot ±10%

Cabled pair orientation (see figure 1).

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TABLE III. Conductor color code.

Pair number	Insulation color
1st pair	Natural/light blue
2nd pair	Natural/light green
3rd pair	Natural/orange

REQUIREMENTS:

Design and construction: (See tables II and III).

Shield location and orientation. The polyester aluminum shield is to be located on the outer circumference of the pair with the aluminum foil side inward. Insulation on the exterior of the shield is to be complete with no aluminum available for contact. The edges of the aluminum foil are to make good contact throughout the length of the pair; the contact surface is to be inward.

Drain wire location. Each drain wire is to be spirally located between the shield and the plastic barrier tape. It is to be in continuous contact with the aluminum surface of the foil shield throughout the cable.

Plastic barrier tape. A polyester tape is to be located between the insulated conductors and the polyester aluminum foil shield.

Cable temperature rating. The cable temperature rating shall be -20°C to +80°C.

Flammability. The cable shall pass UL 1581 VW-1 flame test requirements.

Shield integrity test.*

Frequency	Response **
100 kHz	111 dB below reference
500 kHz	105 dB below reference
1 MHz	99 dB below reference
5 MHz	86 dB below reference
10 MHz	80 dB below reference
15 MHz	75 dB below reference
20 MHz	71 dB below reference
25 MHz	67 dB below reference
30 MHz	64 dB below reference
100 MHz	45 dB below reference

* Any two readings may deviate to a value of 0.9 times the stated limits.

** All values are a minimum numerical value.

Fixture resonance will occur between 40 and 70 MHz.

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Crosstalk test limits. All values are minimums.

Frequency (KHz)	Near end crosstalk (dB)	Far end crosstalk (dB)
40	66	66
70	64	64
100	63	62
1,000	63	60

Insulation resistance between shields: 100 megohms/1,000 feet (minimum).

Durometer hardness. The cable jacket shall have a "Shore A" hardness of 82 ±5.

Part or Identifying Number (PIN): The PIN shall be M49285/45.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Army - MI
Air Force - 71
DLA - ES, IS

Preparing activity:
Army - CR

Agent:
DLA - ES

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